

MATLAB DC Machine Simulator

An interactive MATLAB and Simulink-based project for modeling, analyzing, and simulating the dynamic and steady-state performance of Direct Current (DC) machines.

📌 Project Overview

The **MATLAB DC Machine Simulator** provides an intuitive computational environment to study key electrical and mechanical characteristics—including armature current, back-EMF, speed, electromagnetic torque, and overall efficiency—across various DC machine configurations.

This repository serves as an educational and analytical tool designed for electrical engineering coursework, practical simulations, and research.

Key Features

- **Multiple Machine Operating Modes:**
 - Separately Excited DC Motor
 - DC Shunt Motor
 - DC Series Motor
- **Parameter Sensitivity Analysis:** Observe performance variations under changing load torques and armature resistances.
- **Graphical Visualization:** Automatically generates characteristic curves:
 - Torque vs. Speed
 - Armature Current vs. Torque
 - Efficiency vs. Mechanical Power Output
- **Modular Code Design:** Easily expandable for compound motor configurations or advanced Simulink integration.

🎓 Author & Academic Details

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🛠 Requirements & Dependencies

- **MATLAB** (R2020a or newer recommended)
- **Control System Toolbox** (Optional, for transient response analysis)
- **Simulink** (Optional, if using `.slx` model files)

🚀 Getting Started

1. Clone the Repository

```
``bash
```

```
git clone https://github.com/your-username/MATLAB-DC-Machine-Simulator.git
```

```
``
```

2. Run the Simulation

1. Launch MATLAB.
2. Navigate to the cloned repository directory.
3. Open and run the `main\_simulator.m` script:

```
``matlab
```

```
run('main_simulator.m')
```

```
``
```

📁 Repository Structure

```
``text
```

```
|— README.md      # Project documentation
```

```
|— main_simulator.m # Core MATLAB simulation script
```

```
└— LICENSE       # Open-source license (MIT)
```

```
``
```

License

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